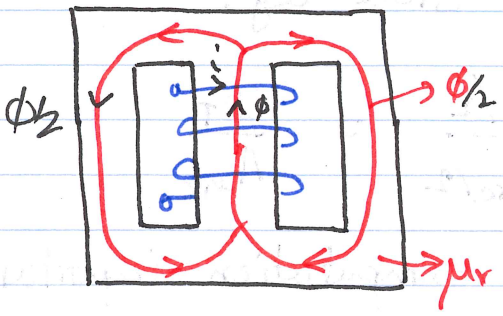
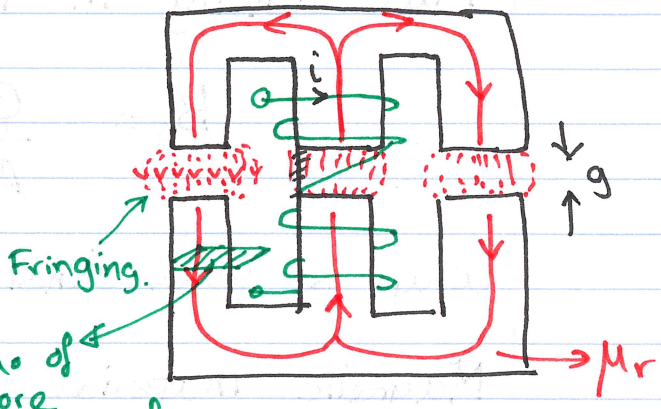
Suggested Reading
: 3.3 - 3.4.1 - 3.4.2

Lecture 10

Last time : * Use magnetic equivalent circuits to calculate inductance.
* Introduction of fringing field.



Ungapped core.



Area of core ≠ area of flux path in the airgap.

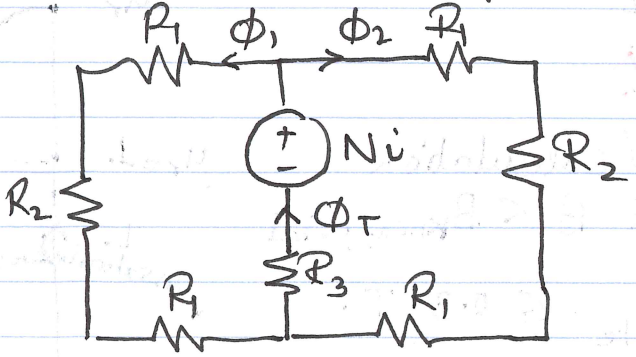
Gapped Core.

Today : ① Computation of flux density (B) $T = \frac{Wb}{m^2}$.

② Concept of flux linkage (λ)

③ Coupling of two coils.

① Flux density, $B = \frac{\Phi}{A}$. $\frac{[Wb]}{[m^2]} = [T]$

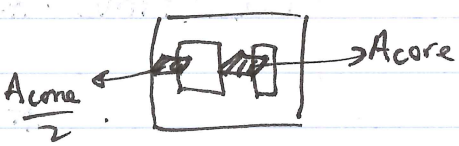


$\Phi_T = \frac{Ni}{R_{eq}} \rightarrow R_{eq} = R_3 + \frac{2R_1 + R_2}{2}$

$\Phi_1 = \frac{\Phi_T}{2} = \Phi_2$

Flux density in the middle leg:

$$B_m = \frac{\Phi_T}{A_{core}}$$



Flux density in the side leg:

$$B_s = \frac{\Phi_T/2}{A_{core}/2} = \frac{\Phi_T}{A_{core}}.$$

Why is flux density computation important?

Magnetic material saturates.

$B = \mu_0 \mu_r H$ ← ideally true.

μ_r: constant

In practice, $B = \mu_0 \mu_r(B) \cdot H$.

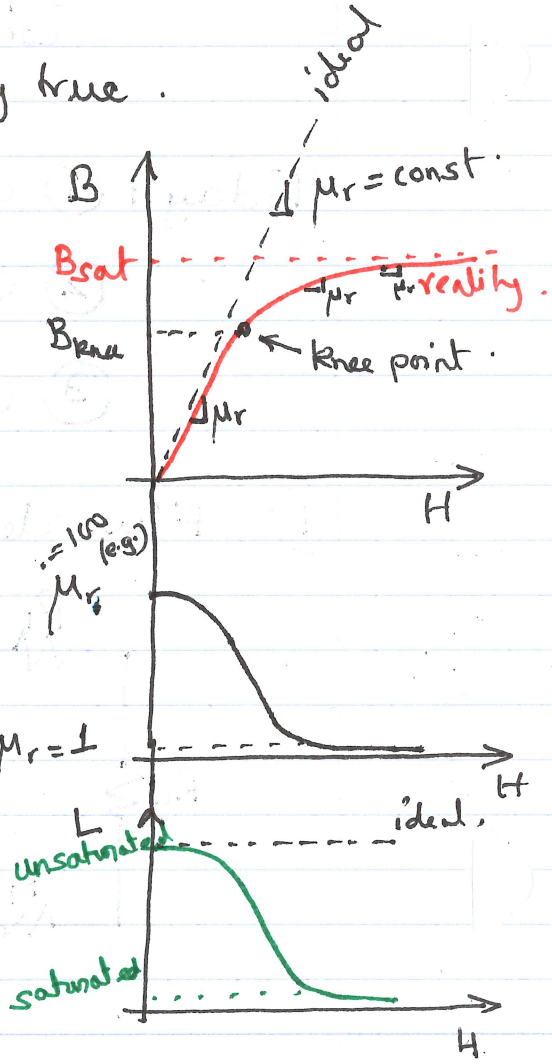
$$R \uparrow = \frac{l}{\mu_0 \mu_r \downarrow H}$$

$$L \downarrow = \frac{N^2}{R \uparrow}$$

Your modeling / calculation is valid until $B < B_{Knee,point}$

$B_{sat}|_{ferrite} \leq 0.3\ T$.

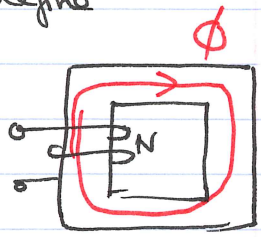
after that $\mu \neq$ constant.



2. Flux linkage:

Mathematically, we will define

$$\lambda = N\phi .$$



$\lambda = N\phi$.

We derived in lecture 8 using Faraday's Law:

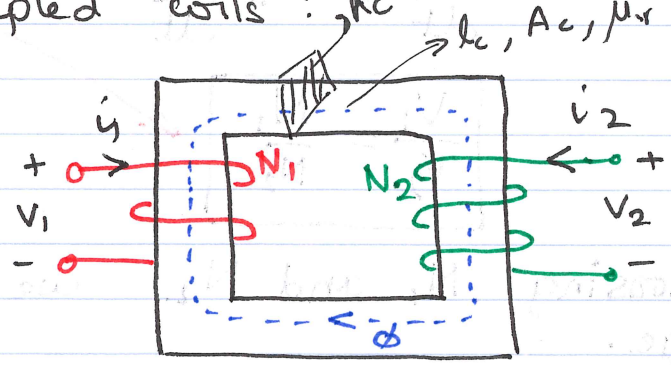
$$\boxed{V = \frac{d}{dt}(N\phi)}$$

$$V = \frac{d\lambda}{dt}$$ ← we will use this later.

Basically, flux linked with N turns.

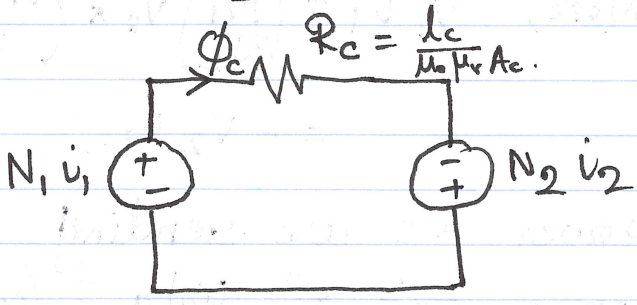
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3. Coupled coils:



Physical system.

Two coils linking a common "flux".



Magnetic equivalent circuit model.

!! Watch out for the polarity of the MMF sources.

Also, windings/coils have no resistance.

Let us make an assumption:
The permeability of the core is infinite.

$$\mu_r \to \infty \qquad R_c = \frac{l_c}{\mu_0 \mu_r A_c} \to 0.$$

$$\therefore \quad N_1 i_1 + N_2 i_2 = \phi_c \cdot R_c \to 0.$$

$$\therefore \quad \boxed{N_1 i_1 + N_2 i_2 = 0}$$

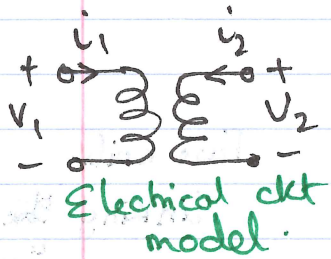
Also,

$$V_1 = \frac{d\lambda_1}{dt} = N_1 \frac{d\phi_c}{dt}$$

$$V_2 = \frac{d\lambda_2}{dt} = N_2 \frac{d\phi_c}{dt}$$

$$\boxed{\frac{V_1}{V_2} = \frac{N_1}{N_2}}$$

Mathematical model



Electrical ckt model.

So, by choosing N_1 and N_2 we can scale voltage.

Note, $V_1 i_1 = \left(\frac{N_1}{N_2}\right) V_2 \times \left(-\frac{N_2}{N_1} i_2\right)$

$= -V_2 i_2$

i.e power we are sending from coil 1. comes out of wdg 2/coil 2.

[lossless].
power is conserved.